

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) — Semester III Examination, 2016-17

Paper No. MTB-301: Algebra (New Course)

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer **five** questions including Question 1 which is compulsory. Figures in right-hand margin indicate marks.*

Question 1. (Compulsory — 2 marks each.)

- (a) If A and B are square matrices of the same order, show that (B^*AB) is Hermitian or Skew-Hermitian according as A is Hermitian or Skew-Hermitian.
- (b) Define rank of a matrix. What is the rank of a 4×4 singular matrix?
- (c) Let G be a group and let $a \in G$. Show that $o(a) = o(a^{-1})$.
- (d) State Lagrange's theorem for finite groups.
- (e) Show that in a group G , if $(ab)^2 = a^2b^2$ for all $a, b \in G$, then G is abelian.
- (f) Show that every group of prime order is cyclic.
- (g) Show that a subgroup of index 2 is a normal subgroup.
- (h) Let $G = \langle a \rangle$ be cyclic of order 13. Find the inverses of a^5 and a^9 and the orders of all four elements.
- (i) Find a regular permutation group isomorphic to $G = \{e, a, a^2, a^3\}$.
- (j) Let $H, K \leq G$ with $|H| = 14$, $|K| = 21$, and $H \cap K \neq \{e\}$. Show $H \cap K \trianglelefteq H$.
- (k) Show that a bijection $f : G \rightarrow G'$ that preserves the operation is an isomorphism.

Question 2.

- (a) Show that every square matrix can be expressed uniquely as $P + Q$ where P is Hermitian and Q is Skew-Hermitian. [6]
- (b) Show that a square matrix A is invertible if and only if $|A| \neq 0$. Further show that if A, B are non-singular of the same order then $(AB)^{-1} = B^{-1}A^{-1}$. [6]

Question 3.

- (a) Reduce the matrix to Echelon form and find its rank: [6]

$$A = \begin{pmatrix} 4 & 2 & 0 & 2 \\ 1 & 2 & 0 & 3 \\ 1 & - & 2 & 1 & 2 \end{pmatrix}$$

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Discuss for all values of λ the system $x + y + 4z = 6$, $x + 2y - 2z = 6$, $\lambda x + y + z = 6$ as regards existence and nature of solutions. [6]

Question 4.

- (a) Prove that a semigroup $(G, *)$ is a group if and only if the equations $ax = b$ and $ya = b$ have solutions in G . [6]
- (b) Prove that a group with four or fewer elements is necessarily abelian. What can you say about groups of orders 5 and 6? [6]

Question 5.

- (a) Define the order of an element of a group.
- (i) Let G be a finite abelian group. Show that $o(ab)$ divides $\text{lcm}(o(a), o(b))$.
- (ii) If $a \in G$ has order n , show $a^m = e$ if and only if $n \mid m$. [6]
- (b) Let $f : G \rightarrow G'$ be a group homomorphism. Show f is injective if and only if $\ker f = \{e\}$. [6]

Question 6.

- (a) Define even and odd permutations. Of the $n!$ permutations on n symbols, show $\frac{n!}{2}$ are even and $\frac{n!}{2}$ are odd. [6]
- (b) Show that the order of a cyclic group equals the order of its generator. Find the number of generators of $U(16)$. [6]

Question 7.

- (a) Prove that $H \trianglelefteq G$ if and only if the product of two right cosets of H in G is again a right coset. [6]
- (b) State and prove the Fundamental Theorem of Homomorphism for groups. [6]

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